

Models to Train Checklist

Legend: ✓ Completed ☐ Pending ☐ To Be Added × Not Implemented

Dataset Classification

ConvNeXt-Tiny

- ✓ Original
- ✓ Depth
- ✓ Canny Edge
- ✓ High-Pass Filter
- ✓ Low-Pass Filter
- ✓ Mean RGB
- ✓ Pixel Shuffle
- ✓ Patch Shuffle (Patch Size 2, 4, 8, 16)
- ✓ Semantic Segmentation
- ✓ VAE
- ✓ Object Detection

SentenceT5

- ✓ Image Captioning

Not Implemented

- × SAM Images
- × Synthetic Images

Gender Artifact Classification

ResNet-50 - Stable Diffusion

- ✓ Original (Full)
- ✓ Full NoBg
- ✓ Mask Segm
- ✓ Mask Rect
- ✓ Mask Segm NoBg
- ✓ Mask Rect NoBg
- ☐ Downsampled Color (112×112, 56×56, 28×28, 14×14, 7×7)
- ☐ Downsampled Grayscale (112×112, 56×56, 28×28, 14×14, 7×7)

Logistic Regression

- ✓ Mean RGB
- ✓ One-Hot Encoded Objects (COCO)
- ✓ Depth at Keypoints (*Novel Extension*)

MLP

- ☐ Pose (Keypoints)

ResNet-50 - ReLaion5B

- ✓ Original (Full)
- ☐ Full NoBg
- ☐ Mask Segm
- ☐ Mask Rect
- ☐ Mask Segm NoBg
- ☐ Mask Rect NoBg
- ☐ Downsampled Color (112×112, 56×56, 28×28, 14×14, 7×7)
- ☐ Downsampled Grayscale (112×112, 56×56, 28×28, 14×14, 7×7)

Logistic Regression

- ☐ Mean RGB
- ☐ One-Hot Encoded Objects (COCO)
- ☐ Depth at Keypoints (*Novel Extension*)

MLP

- ☐ Pose (Keypoints)

Skin Tone Artifact Classification

ResNet-50 Stable Diffusion

- ☐ Original (Full)
- ☐ Full NoBg
- ☐ Mask Segm
- ☐ Mask Rect
- ☐ Mask Segm NoBg
- ☐ Mask Rect NoBg
- ☐ Downsampled Color (112×112, 56×56, 28×28, 14×14, 7×7)
- ☐ Downsampled Grayscale (112×112, 56×56, 28×28, 14×14, 7×7)

Logistic Regression

- ☐ Mean RGB
- ☐ One-Hot Encoded Objects (COCO)
- ☐ Depth at Keypoints (*Novel Extension*)

MLP

- ☐ Pose (Keypoints)

ResNet-50 ReLaion5B

- ☐ Original (Full)
- ☐ Full NoBg
- ☐ Mask Segm
- ☐ Mask Rect
- ☐ Mask Segm NoBg
- ☐ Mask Rect NoBg
- ☐ Downsampled Color (112×112, 56×56, 28×28, 14×14, 7×7)
- ☐ Downsampled Grayscale (112×112, 56×56, 28×28, 14×14, 7×7)

Logistic Regression

- ☐ Mean RGB
- ☐ One-Hot Encoded Objects (COCO)
- ☐ Depth at Keypoints (*Novel Extension*)

MLP

- ☐ Pose (Keypoints)

Spurious Feature Analysis (adapted from Zeng et al.)

ResNet-50 (x4 Stable Diffusion / ReLaion5B & Gender / SkinTone)

- ☐ Canny Edge
- ☐ High-Pass Filter
- ☐ Low-Pass Filter
- ☐ Pixel Shuffle
- ☐ Patch Shuffle (Patch Size 2, 4, 8, 16)
- ☐ Semantic Segmentation
- ☐ VAE
- ☐ Object Detection

SentenceT5

- ☐ Image Captioning

Extra (Time Permitting) — Re-LAION-5B Source Website Classification

ConvNeXt-Tiny

- ☐ Original
- ☐ Depth
- ☐ Canny Edge
- ☐ High-Pass Filter
- ☐ Low-Pass Filter
- ☐ Mean RGB
- ☐ Pixel Shuffle
- ☐ Patch Shuffle (Patch Size 2, 4, 8, 16)
- ☐ Semantic Segmentation
- ☐ VAE
- ☐ Object Detection

SentenceT5

- ☐ Image Captioning